C 22569 (Pages : 3) Name.....

Reg. No.....

FOURTH SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, APRIL 2022

April 2021 Session for SDE/Private Students

(CBCSS)

Mathematics

MTH 4E 09—DIFFERENTIAL GEOMETRY

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

General Instructions

Covid Instructions are not applicable for Pvt/SDE students (April 2021 session)

1. *In cases where choices are provided, students can attend* **all** *questions in each section.*
2. *The minimum number of questions to be attended from the Section/Part shall remain the same.*
3. *The instruction if any, to attend a minimum number of questions from each sub section/sub part/sub division may be ignored.*
4. *There will be an overall ceiling for each Section/Part that is equivalent to the maximum weightage of the Section/Part.*

Part A

Answer **all** *questions.*
Each question carries a weightage 1.

1. Sketch the vector field X on $\mathbb{R}^2$ defined by $\bar{X}(p) = (p, X(p))$ where $X(p) = X(x_1, x_2) = (-x_2, x_1)$.
2. Show that graph of any function $f : \mathbb{R}^n \to \mathbb{R}$ is a level set for some function $F : \mathbb{R}^{n+1} \to \mathbb{R}$.
3. Define orientation on the n-surface. Give an example of a non-orientable surface.
4. Define Geodesic. Show that geodesic have constant speed.
5. Define Levi-Civita parallel vector field. Show that if X and Y are Levi-Civita parallel vector fields along α, X. Y is constant along α.
6. Define Weingartan map. Why it is called sphere operator ?

Turn over

7. Define parameterized n-surface in $\mathbb{R}^{n+k}, k \geq 0$.

8. Define (i) Differential 1-form ; (ii) Exact 1-form.

(8 × 1 = 8 weightage)

Part B

Answer **two** *questions from each unit.*
Each question carries a weightage 2.

Unit I

9. Find the integral curve through P (1, 0) and P(a, b) for the vector field $\overline{\mathrm{X}}(p) = (x_1, x_2, 0, 1)$ on $\mathrm{U} \subseteq \mathbb{R}^2$.

10. Let S be an n-surface in $\mathbb{R}^{n+1}, g : \mathrm{U} \to \mathbb{R}$ is a smooth function and $p \in \mathrm{S}$ is an extreme point of g on S. Then show that there exist a real number λ such that $\nabla g(p) = \lambda \nabla f(p)$.

11. Let $\mathrm{S} \subset \mathbb{R}^{n+1}$ be a connected n-surface $\mathbb{R}^{n+1}$. Then show that there exist on S exactly two smooth unit normal vector fields $\overline{\mathrm{N}}_1$ and $\overline{\mathrm{N}}_2$ and $\overline{\mathrm{N}}_2(p) = -\overline{\mathrm{N}}_1(p) \forall p \in \mathrm{S}$.

Unit II

12. Show that a parametrized curve α in the unit n-sphere $x_1^2 + \ldots + x_{n+1}^2 = 1$ is a geodesic if and only if it is of the form $\alpha(t) = (\cos at)\overline{e}_1 + (\sin at)\overline{e}_2$ for some orthogonal pair of unit vectors $\{e_1, e_2\} \in \mathbb{R}^{n+1}$ and some $\alpha \in \mathbb{R}$.

13. Let S be an n-surface in $\mathbb{R}^{n+1}$, let $\alpha : \mathrm{I} \to \mathrm{S}$ be a parametrized curve in S, let $t_0 \in \mathrm{I}$ and let $v \in \mathrm{S}_{\alpha(t_0)}$. Then show that there exists a unique vector field V, tangent to S along α, which is parallel and has $\mathrm{V}(t_0) = v$.

14. Evaluate the Weingarten map L_p for $x_2^2 + x_3^2 = a^2$ in $\mathbb{R}^3$.

Unit III

15. Find normal curvature of the n-plane $x_1 + x_2 + \ldots + x_{n+1} = 1$ at p = (1, 0, …, 0) oriented by $\overline{\mathrm{N}} = \nabla f / \|\nabla f\|$.

16. Prove that if V is a finite dimensional vector space with dot product and L : V → V a self-adjoint linear transformation on V. Thent there exists an orthonormal basis for V consisting of eign vectors of L.

17. Find the Gauss-Kronecker curvature of a cylinder over an n-surface.

(6 × 2 = 12 weightage)

Part C

Answer any **two** *questions.*
Each question carries a weightage of 5.

18. (a) Sketch the Graph f and level set of the function $f(x_1, x_2) = x_1$.

    (b) Let U be an open set in $\mathbb{R}^{n+1}$ and let $f : U \to \mathbb{R}$ be smooth. Let $p \in U$ be a regular point of f and $c = f(p)$. Then show that the set of all vectors tangent to $f^{-1}(c)$ at p is equal to $[\nabla f(p)]^{\perp}$.

19. Let C be an oriented plane curve. Then there exists a global parametrization of C if and only if C is connected.

20. Let η be the 1-form on $\mathbb{R}^2 - \{0\}$ defined by $\eta = \frac{-x_2}{x_1^2 + x_2^2} dx_1 + \frac{x_1}{x_1^2 + x_2^2} dx_2$. Prove that for $\alpha : [a, b] \to \mathbb{R}^2 - \{0\}$ any closed piecewise smooth parametrized curve in $\mathbb{R}^2 - \{0\}$, $\int_{\alpha} \eta = 2\pi k$ for some integer k.

21. State and prove inverse function theorem for n-surface.

(2 × 5 = 10 weightage)